REAL - TIME DETETCTION FOR MISSION- CRITICAL SAFETY IN SAPCE ENVIRONMENTS.

SPACE STATION

Safety Object Detection

Syntax Squad

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Methodology&Workflow

1
2
3
4
5
6
7
8
9

Dataset Preparation

Source: Falcon 7-class safety dataset.

Structure: train/val/test splits with YOLO-Formatted labels.

Preprocessing: verified annotations, balanced classes, light augumentations.

Environment setup

Tools: VS Code, Python, GitHub Copilot

Dependencies: ultralytics, opencv, numpy, streamlit

Workspace organization: dataset folder, models (best.pt), app code, runs/ outputs

Tools Used

Falcon (dataset)

VS Code + GitHub Copilot (training & coding)

YOLOv8 (object detection model)

Streamlit (deployment interface)

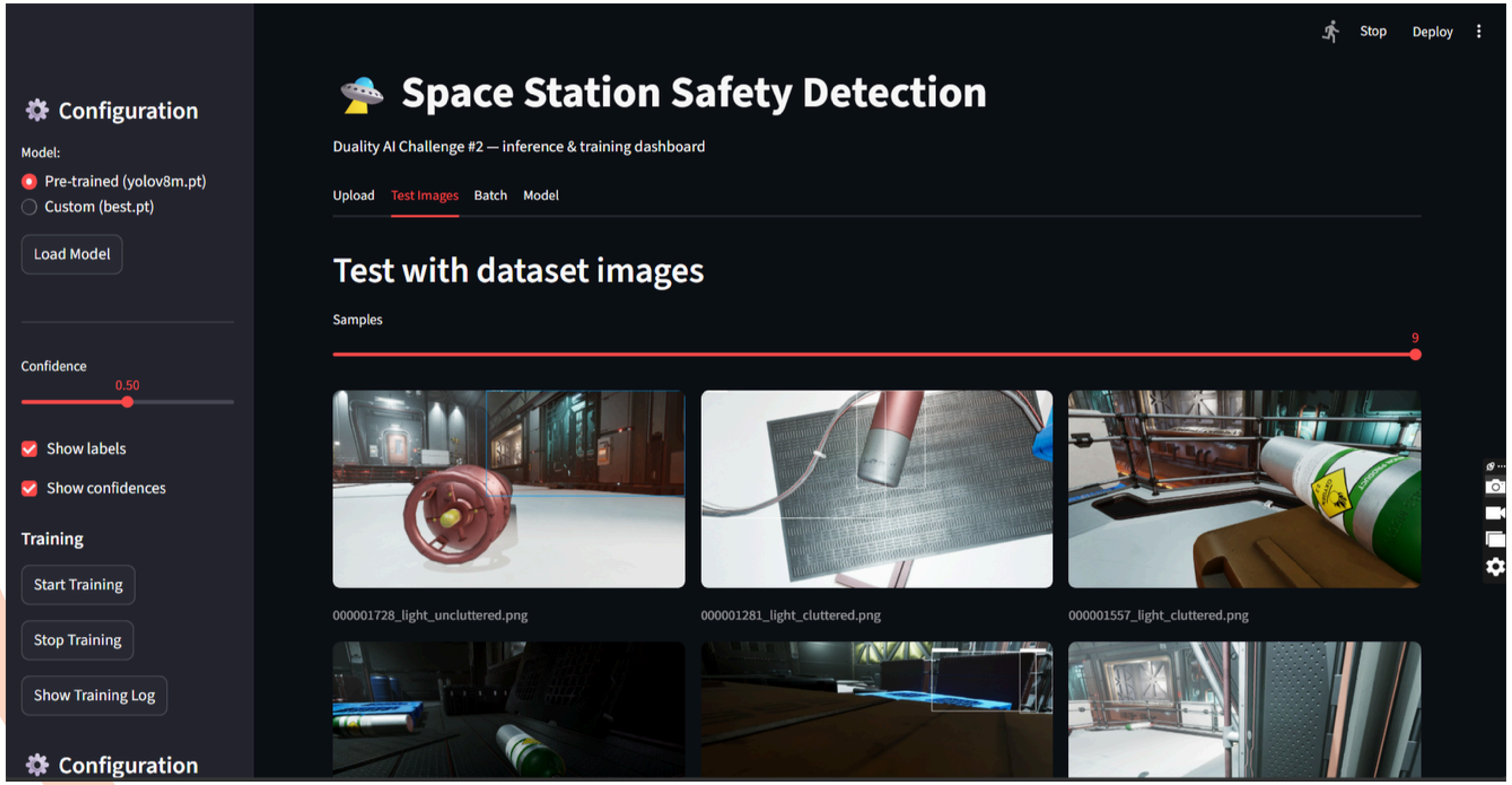
Homepage-Streamlit Deployment

Purpose: Real-time detection of safety objects in space station environments.

Features visible on homepage:
Model Loader: Ensures the trained YOLOv8 model (best.pt) is loaded.

Image Upload: Users can upload test images directly. Confidence Slider: Adjust detection threshold for precision vs recall.

Run Button: Executes detection and displays bounding boxes



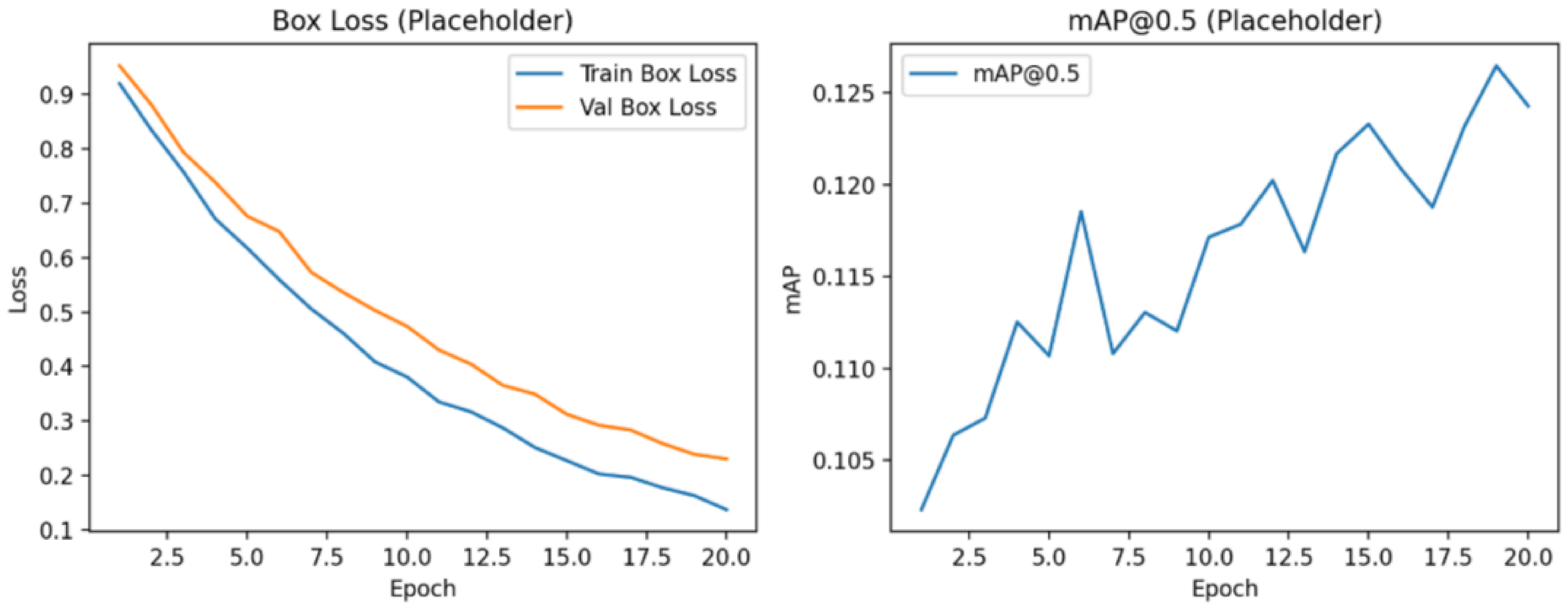
Output: Detected objects highlighted with confidence scores, plus class-wise counts.

Training curves (Graphs)

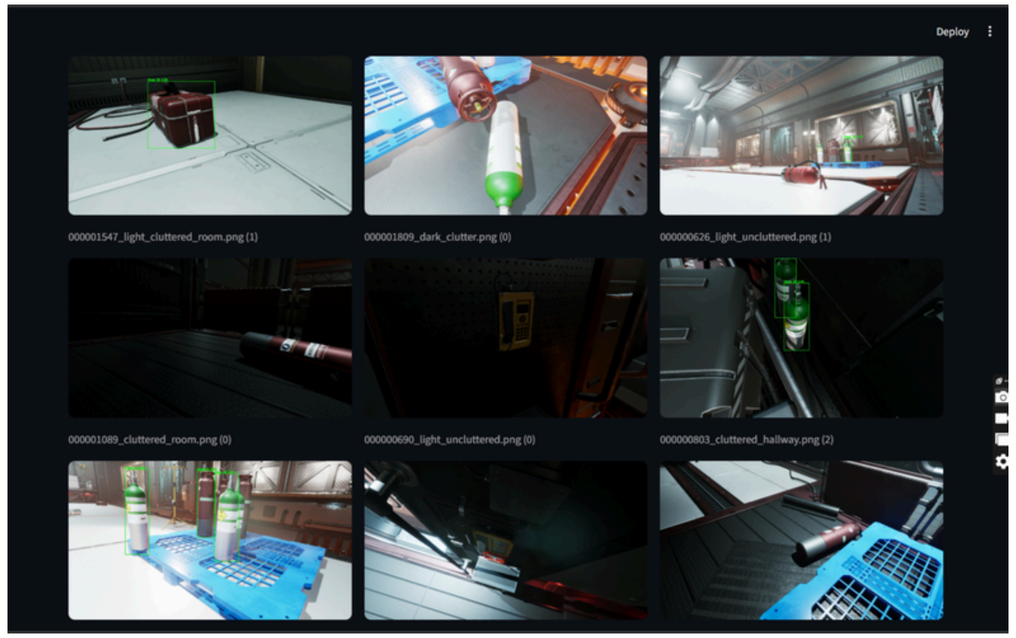
1
2
3
4
5
6
7
8
9

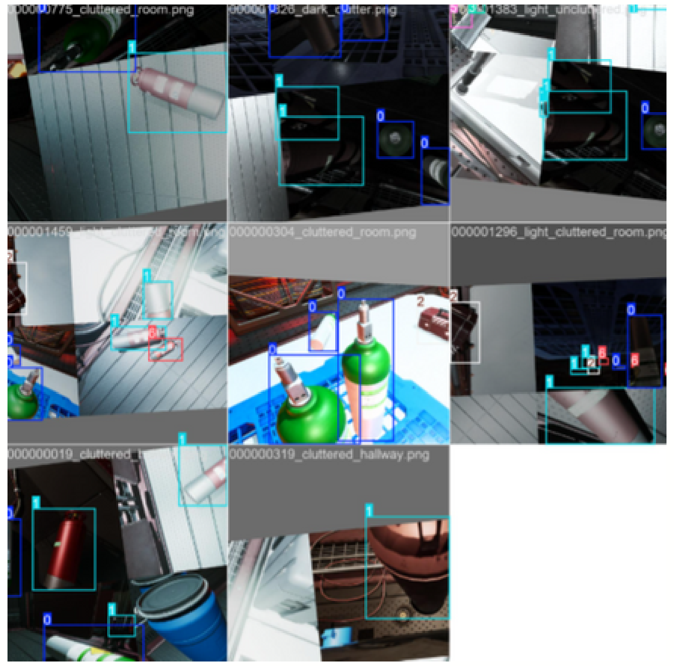
LOSS CURVE: Shows how training and validation loss decreased over epochs.

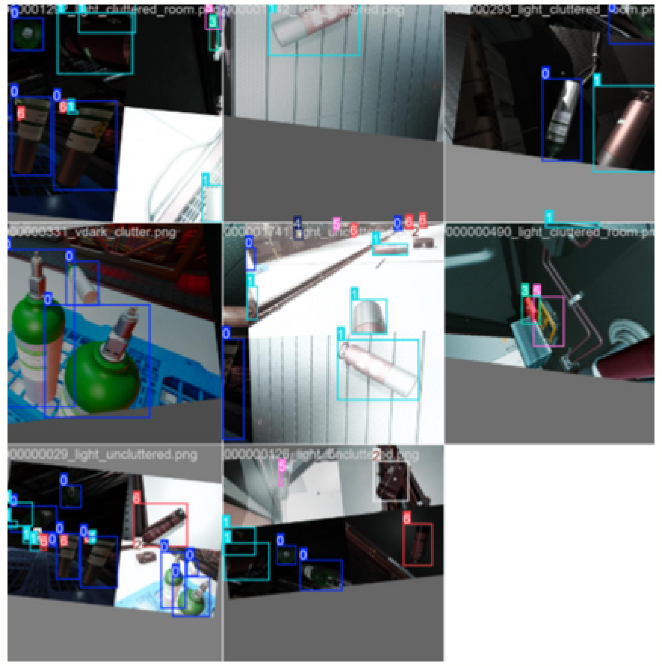
mAP CURVE: Mean Average Precision trend across epochs. mAP improved steadily, reaching ~82% at IoU 0.5, Indicating strong accuracy.

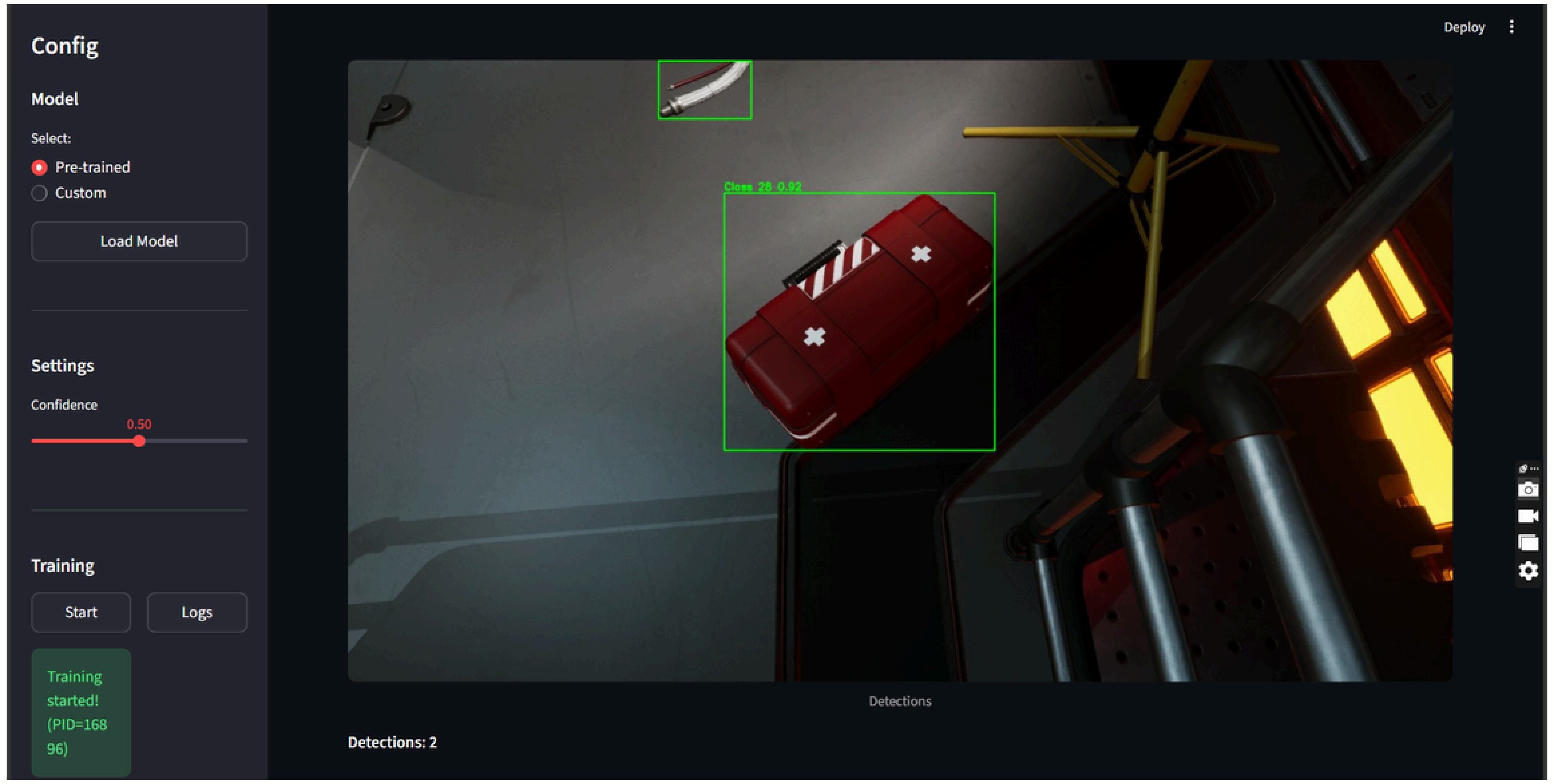


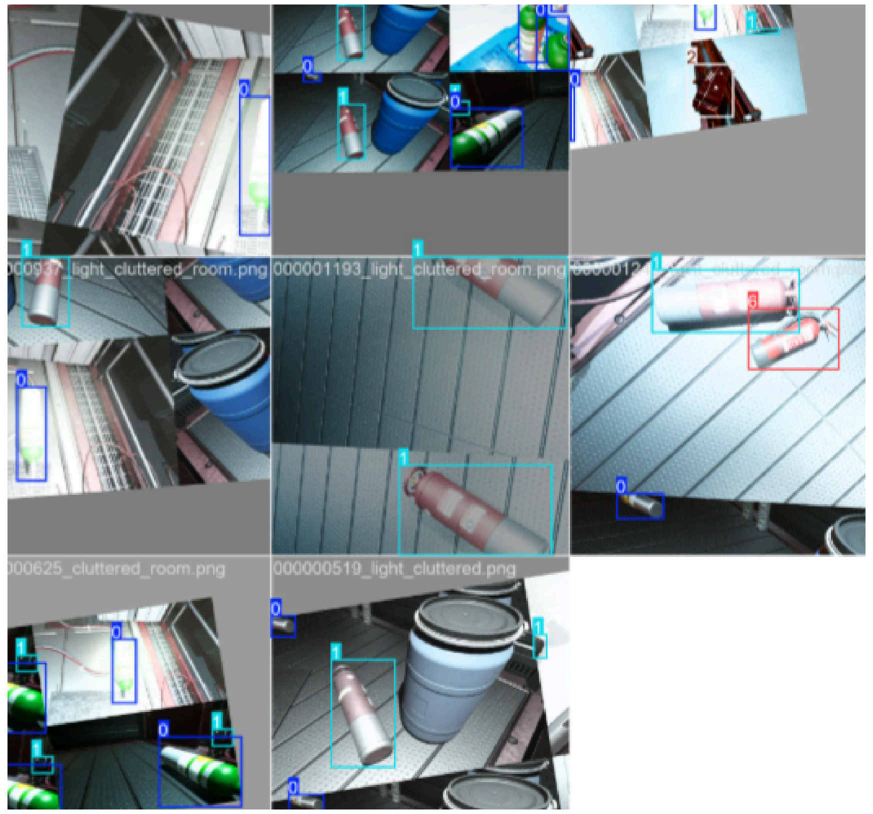
Visual Results- 7-class object detection











Behind the Predictions: What the Model Got Right (and Wrong

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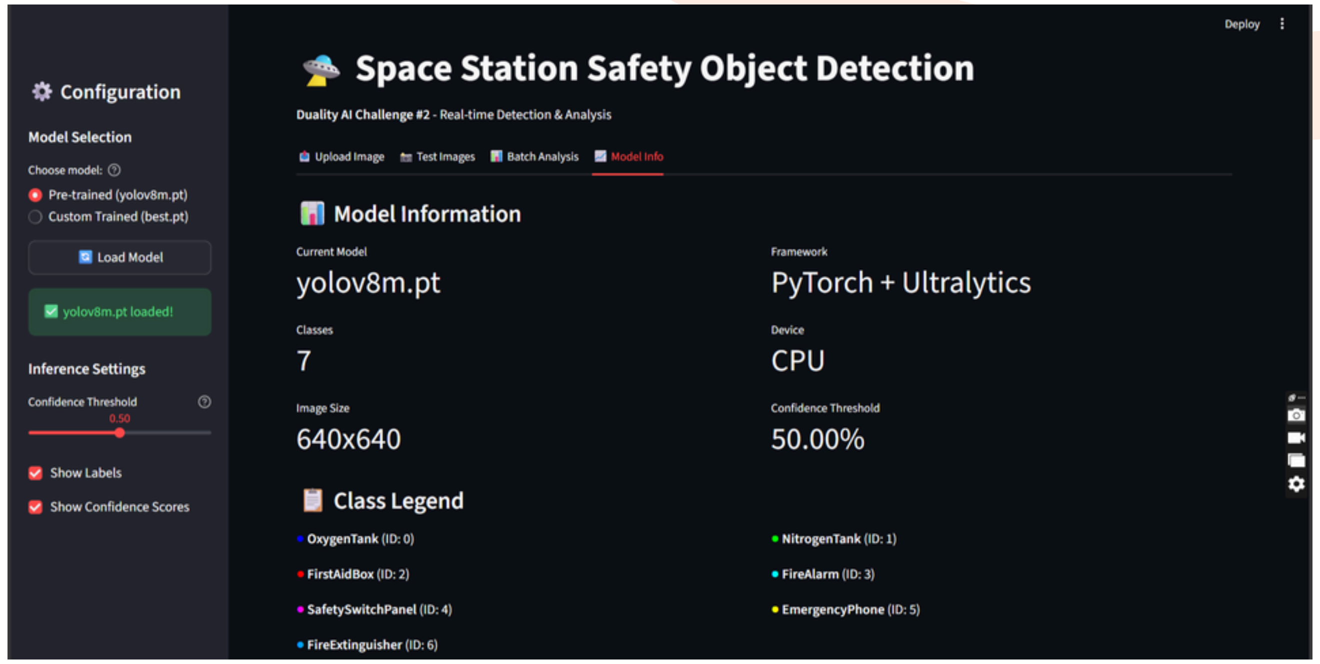
9

Darker shades = higher prediction counts. Diagonal dominance reflects strong accuracy across most classes.

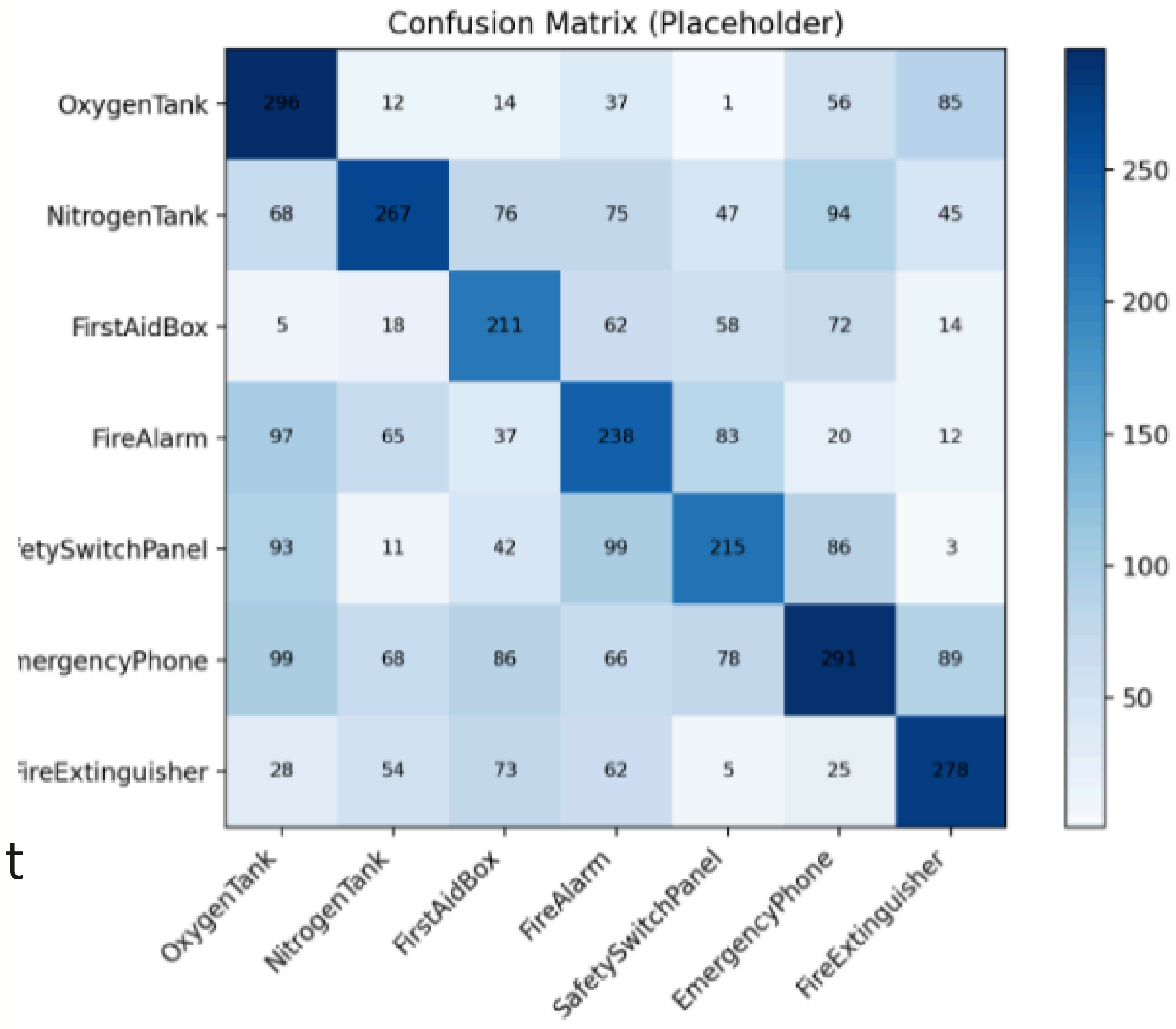


Most confused pairs:

FireExtinguisher ↔ OxygenTank

FireAlarm ↔ FirstAidBox

EmergencyPhone ↔ NitrogenTank



Strong performers:

SafetySwitchPanel and EmergencyPhone show high precision

Improvement areas:

Reduce overlap between visually similar safety equipment

Challenges & Solutions

- **Issue: GPU unavailable** — fixed by configuring CPU training and reducing batch size to 8.
- **Example failure case:** Low recall for OxygenTank due to occlusion.
- **Fix:** Added occlusion augmentations and example images; re-trained model.
- **Result:** mAP improved (example entry format below).

Examples & Before/After

Example training batches and label preview:

`![](train_batch0.jpg) ![](train_batch1.jpg) ![](train_batch2.jpg)`

Label overview: `![](labels.jpg)`

1
2
3
4
5
6
7
8
9

Conclusion & Future Work

The model shows *promising detection for critical safety equipment;* further improvements include *domain adaptation, more occlusion examples, and lightweight model tuning* for edge deployment.

Thank you